

free for use. Anyone can use a public blockchain network.

Private: Private blockchains exist within a single organisation, allowing only specific people of the organisation to verify and add transaction blocks. However, everyone on the Internet is generally allowed to view.

Consortium: In this blockchain variant, only a group of organisations can verify and add transactions. Here, the ledger can be open or restricted to select groups. A consortium blockchain is used across organisations. It is only controlled by pre-authorised nodes.

Pillars of blockchain platforms

When there are many public and private blockchain platforms available in the marketplace, one has to carefully choose a suitable option based on one's requirements. This has to be evaluated based on the pillars of the blockchain platform as listed below:

- Decentralised network
- Platform security
- Immutability of the record state

Decentralised network: One of the key architectural principals of the blockchain platform is its decentralised nature. This means the transaction in the blockchain network is copied across all the nodes of the network

and all the nodes are connected. This blockchain platform is highly reliable as there is no possibility of tampering with the transaction record because it is copied to all the nodes — and tampering with all the nodes in the network is practically impossible at any given time.

Platform security: Though the blockchain platform is decentralised in nature with many users being a part of the workflow process and participating in the various stages of executing the transaction, there is a higher level of security ensured in any blockchain platform due to its decentralised nature and multi-node record copy. Also, different blockchain platforms ensure higher levels of platform security like permissionless ledgers, the consensus algorithm, usage of cryptocurrency for transactions and the smart contract facility, to name a few.

Record immutability: In any blockchain platform, the decentralised network and ledger copied across all the nodes of the network ensures that all the records kept in the ledger for any transaction are secure enough so that a change in a record is accepted only if it's accepted by all participants across nodes, thus changing the record unanimously across all the ledger

copies in the network. This ensures that the records in the transaction processing stages are immutable in nature. Any correction in the record is amended as a new transaction processing stage in the network, thus ensuring all the record processing transaction stages are kept as distinct entries in the ledger.

Blockchain platforms

Blockchain platforms assist us in creating applications that implement the concepts of blockchain. Every person or company does not have either the resources or the time to develop their own blockchain from scratch, and hence these companies leverage blockchain platforms developed by tech giants for faster and easier application development. Any organisation deciding to implement blockchain may choose from the prominent frameworks available, i.e., Ethereum, Hyperledger Fabric, Quorum, Corda, Ripple, etc. The final decision should be based upon the suitability of each, to the organisation. Let's now discuss the most popular open source blockchain platforms.

Ethereum

Ethereum (a public blockchain network) was developed by Vitalik Buterin and is considered an efficiently developed platform that has smart contract features, flexibility and multi-industry adaptability. Ethereum acts as a base component in building and developing most of the decentralised applications. ERC20 is the most popular token standard among cryptocurrencies. Stability, security, corruption prevention and zero downtime are some of the benefits offered by Ethereum over other applications.

The Ethereum network has two types of accounts, namely:

- External accounts
- Contract accounts

Both are referred to as 'state objects' and comprise the 'state' of the Ethereum network, i.e., every state object has a well-defined state. For

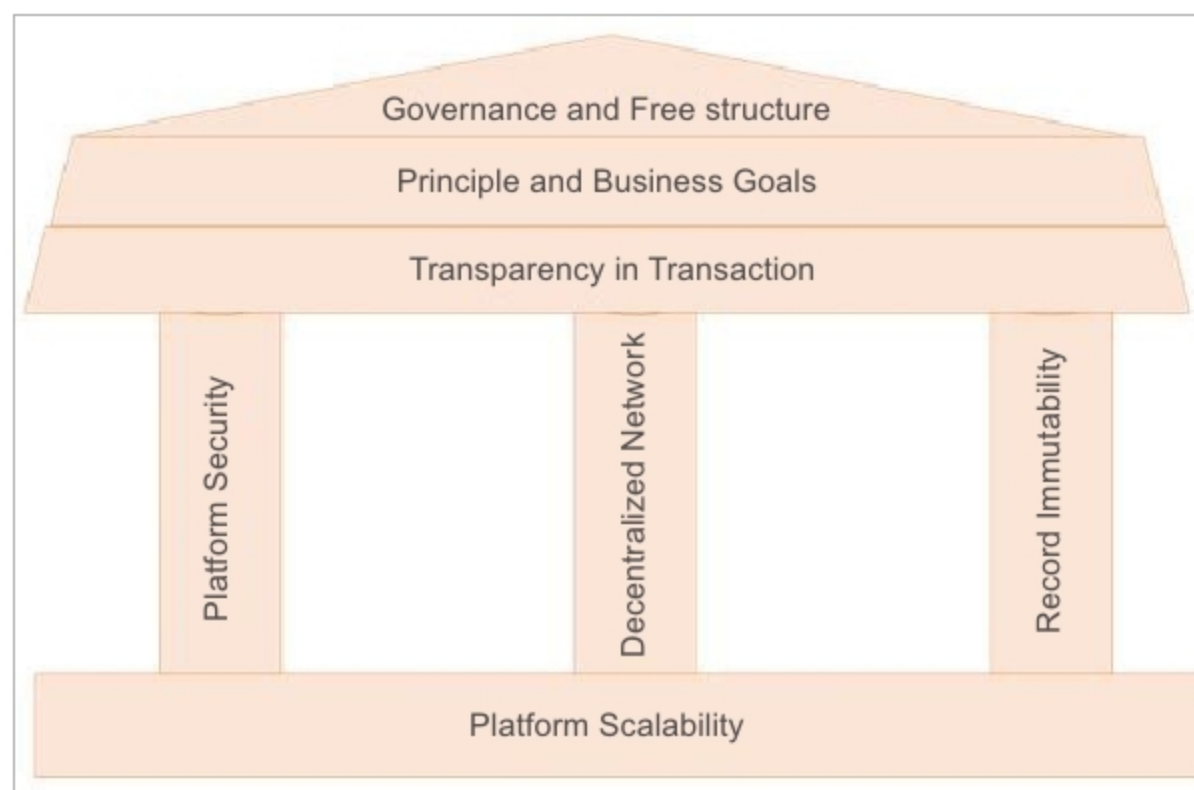


Figure 1: Pillars of blockchain platforms